Boundary-Aware Token-Structure Analysis of the Voynich Manuscript:
A Reproducible Reassessment

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Abstract

The Voynich Manuscript (Beinecke MS 408) is a fifteenth-century codex written in an undeciphered script. We analyze token-family transitions and automatically discovered prefix and suffix families in the frozen Zandbergen–Landini EVA transcription (31608 tokens in 4197 lines on 184 pages). This revision replaces earlier analyses that flattened sequence boundaries or used unmatched comparator sizes. Under a canonical prefix-first classifier and strictly within Voynich lines, CHEDY→QOK occurs 615 times, or 2.659× its independence expectation, while AIIN→QOK occurs 127 times, or 0.444× expectation. These are descriptive transition-matrix effects; matrix-wide cluster-aware inference remains open. Raw `aiin`/`ain` substring density has similar observed means in Currier A and B pages, but equivalence was not tested and canonical AIIN-family density differs between the groups.

The central cross-corpus analysis applies identical automatic affix discovery to Voynich and every comparator, preserves lines and sentences, resamples Voynich by page blocks, and repeatedly samples comparator sequence units without replacement to exactly 31608 tokens. Across 200 replicates, Voynich has median prefix self-clustering 1.333 (95% replicate interval $[1.222, 1.461]$), suffix self-clustering 1.458 ($[1.389, 1.555]$), and ratio 0.918. Its median minimum-side score is 1.333. Including the catch-all `OTHER` class lowers the corresponding medians to 1.292, 1.363, and 0.951. Voynich exceeds each of 14 eligible comparators on the continuous minimum-side score in these sampled distributions; corrected two-sided replicate sign tests are $p = 0.00995$ and remain 0.00995 after Benjamini–Hochberg adjustment. This finite-set result does not establish natural-language uniqueness, language identity, decipherment, or exclusion of constructed, hybrid, stenographic, cipher, or other historical mechanisms. One declared comparator, Ottoman Turkish, is ineligible because its preserved sentence corpus contains only 16890 usable tokens and is not padded.

Keywords: Voynich manuscript, corpus linguistics, sequence boundaries, self-clustering, reproducible research, statistical text analysis.

Scope statement. This paper measures properties of one transcription under declared procedures. It does not translate the manuscript, assign semantic meanings, identify a language, prove natural or encoded language, or rule out untested mechanisms.

1 Introduction

The Voynich Manuscript is a codex of unknown authorship held by Yale's Beinecke Rare Book and Manuscript Library as MS 408 (Clemens, 2016). Radiocarbon dating places its vellum between 1404 and

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1438 (Hodgins, 2011). Numerous decipherment proposals have been advanced, but none has achieved independent verification. A separate line of work treats the transcription as a corpus and studies its statistical structure without attempting translation. Prior work has reported non-randomness, Zipf-like frequency structure, and section-associated vocabulary (Reddy and Knight, 2011; Montemurro and Zanette, 2013; Bowern and Lindemann, 2021).

This project belongs to the latter tradition. It asks two limited questions. First, how do five declared EVA token families transition within manuscript lines? Second, when prefix and suffix families are discovered by the same algorithm in every corpus, how do their self-clustering estimates compare in a fixed set of size-matched corpora?

The present revision follows a technical audit that identified conflicts among historical scripts, results, tests, and prose. Earlier public versions used flattened token streams in the prefix/suffix analysis, allowed artificial adjacency at corpus and resampling seams, used unequal corpus sizes, included `OTHER` in the headline mean, and applied a hand-selected Voynich prefix partition while discovering comparator families automatically. Those results, including the values 1.52, 1.54, and ratio 0.99, are retired. The historical files remain archived in the repository rather than being silently overwritten.

2 Data and Methods

2.1 Frozen Voynich corpus

The canonical input is the Zandbergen–Landini source within the frozen `AncientLanguages/Voynich` parquet snapshot. Canonical tokenization produces 31608 tokens in 4197 nonempty lines on 184 pages. SHA-256 hashes for the input and relevant code are recorded in the generated result. Lines are the natural adjacency units. Pages are the block units for Voynich uncertainty resampling. No canonical transition is constructed across lines, pages, sections, hands, removed-token positions, or resampled blocks.

2.2 Declared EVA families and ambiguity

The core transition analysis uses five predicates plus `OTHER`:

- QOK: token begins with `qok`;
- OK: token begins with `ok` but not `qok`;
- OT: token begins with `ot`;
- CHEDY: token contains `chedy`, `shedy`, `chey`, or `shey`;
- AIIN: token contains `aiin` or `ain`.

These predicates overlap. Exactly 1,423 token instances (4.5020%; 160 types) match more than one family. The canonical policy gives prefix predicates precedence in the order QOK, OK, OT, CHEDY, AIIN. A strict sensitivity policy removes ambiguous tokens and breaks the sequence at every removed position, so former neighbors never become adjacent. The predicates are EVA-specific and are not paleographic identifications.

2.3 Within-line transition estimates

For a source class s and destination class d, the descriptive ratio is

$$R(s \to d) = \frac{N_{s\to d}}{N_s^{\mathrm{src}} N_d^{\mathrm{dst}} / N_{\mathrm{transitions}}}.$$

Counts and marginals are pooled only over transitions that already occur within separate lines. The permutation null shuffles class labels within each line, uses 2,000 replicates, and reports the corrected empirical estimate $(b+1)/(B+1)$ rather than zero. Individual transition cells remain clustered within lines and pages, so the cell-level permutation values and the ordinary 6×6 independence statistic are treated as descriptive pending a matrix-wide cluster-aware analysis.

2.4 AIIN density decomposition

Two quantities are kept distinct. Raw substring density counts every token containing `aiin` or `ain`. Canonical-family density counts only tokens assigned to AIIN after precedence resolves overlap. Per-page Currier A and B distributions are compared with a two-sample Kolmogorov–Smirnov test. A nonsignificant difference test is not interpreted as proof of equivalence or invariance.

2.5 Comparator corpora and sequence units

The Version 2 comparison declares 12 Leipzig Wikipedia corpora: Arabic, Latin, Estonian, Hebrew, Finnish, Hungarian, Turkish, Italian, North Azerbaijani, Swahili, Georgian, and Tagalog. Each Leipzig sentence record is one sequence unit. Middle English *Canterbury Tales* and KJV English are frozen Gutenberg documents split into sentence-like units after removing Gutenberg headers and footers. Ottoman Turkish is read from genuine CoNLL-U sentence boundaries. Mandarin and pending corpora are not part of this estimator.

Leipzig texts are modern Wikipedia proxies, not historical or genre-matched controls. The two Gutenberg texts are historical literary comparators, not matched manuscript genres. Ottoman Turkish provides only 16890 usable tokens under preserved sentence boundaries, below the target 31608, and is therefore declared ineligible. It is neither flattened nor padded.

2.6 Matched resampling

Each eligible comparator replicate randomly orders natural sequence units without replacement and takes whole units until reaching the exact target of 31608 tokens. If the final unit would overshoot, a random contiguous fragment supplies the remaining tokens and remains a separate sequence. Voynich uncertainty is estimated with page-block bootstrap replicates. A sampled page contributes its original separate lines; a final partial line is a separate contiguous fragment. These procedures cannot create adjacency across sentence, document, line, page, sample, or truncation boundaries.

The analysis uses 200 replicates and master seed `20260718`. Every child seed is recorded. Replicate intervals are percentile summaries of the declared resampling distributions, not guarantees about an unobserved population of all languages or transcriptions.

2.7 Symmetric affix discovery and self-clustering

For each system and replicate, the same discovery procedure is applied separately to prefixes and suffixes. Candidate affixes have length two or three code points, cover 2% through 20% of tokens, and are selected from the 80 most frequent candidates. Selection is greedy with lexical tie-breaking; nested candidates on the same side are excluded; at most five families are retained. Each token is assigned to the first selected family it matches or to `OTHER`.

For discovered class c, self-clustering is observed within-unit $c \to c$ divided by the expectation from pooled within-unit source and destination marginals. A class enters the unweighted mean when both marginals exceed 10 and its expected self-transition count exceeds one. The primary score excludes `OTHER`. Including `OTHER` is a required sensitivity condition. Primary continuous outcomes are prefix score P, suffix score S, ratio P/S, and balanced elevation $\min(P,S)$.

Bucket labels are secondary. Threshold sensitivity crosses elevation thresholds 1.05, 1.10, and 1.15; ratio lower bounds 0.75, 0.80, and 0.85; and upper bounds 1.20, 1.25, and 1.30, for 27 cells.

For each eligible comparator, replicate-wise differences in $\min(P,S)$ are summarized by their median and percentile interval. A corrected two-sided replicate sign estimate is reported and adjusted by

Benjamini–Hochberg over the 14 eligible comparator tests. The smallest possible two-sided corrected value with 200 replicates is $2/(200+1) = 0.00995$.

2.8 Reproducibility and source hierarchy

The canonical hierarchy is shared code, frozen inputs and manifests, pipeline-listed scripts, fresh generated results, tests, current prose, the paper, and finally archived analyses. Generated outputs override prose. The command `python run_all.py` validates inputs, regenerates the core and prefix/suffix analyses, runs pytest and direct Python execution, checks output freshness, and writes a run manifest. Estimator version, command, input and code hashes, Git revision, and seeds are stored in `results/prefix_suffix_v2.json`.

3 Results

3.1 Within-line transition structure

Under canonical precedence and strictly within lines, CHEDY→QOK has 615 observations against an expectation of approximately 231, giving $R = 2.659$. AIIN→QOK has 127 observations against an expectation of approximately 286, giving $R = 0.444$. Both corrected within-line shuffle values are $p = 0.0005$. CHEDY→QOK spans 312 unique canonical within-line token pairs. Among 58 sufficiently frequent CHEDY token types, 44 (76%) meet the project's declared attraction criterion.

These estimates establish class-level departures from the independence baseline in this representation. They do not by themselves establish syntax, semantics, or independence of transition events. The ordinary matrix statistic is therefore not used to make individual confirmatory cell claims.

3.2 AIIN is not canonically invariant

Raw `aiin`/`ain` substring density has observed page means of approximately 15.0% in both Currier A ($n = 102$ pages) and B ($n = 72$), with KS $p = 0.742$. The difference procedure did not reject equality, but no equivalence margin or equivalence test was specified. The appropriate result is descriptive similarity, not invariance.

After canonical precedence resolves overlapping families, mean AIIN-family density is 13.58% in Currier A and 10.77% in Currier B, with KS $p = 0.0028$. Thus the earlier broad claim that AIIN-family density is invariant is rejected. Raw substring occurrence and canonical assignment answer different questions and must not be substituted for one another.

3.3 Boundary-aware prefix/suffix comparison

Table 1 reports primary medians. Voynich has prefix score 1.333, suffix score 1.458, ratio 0.918, and minimum-side score 1.333. The prefix and suffix 95% replicate intervals are $[1.222, 1.461]$ and $[1.389, 1.555]$, respectively.

All 14 primary Δmin distributions are positive at their reported 2.5th percentiles. Each corrected two-sided replicate sign estimate is $p = 0.00995$; because all raw values tie, Benjamini–Hochberg adjustment leaves them at 0.00995. This resolution is determined by the 200-replicate design. It should not be reported as stronger evidence than the design permits.

The minimum-side result is not equivalent to saying that both Voynich side scores exceed both comparator side scores. For example, Latin has much higher suffix clustering, and Arabic has higher median prefix clustering. The result instead says that Voynich has the larger weaker-side score in the finite tested set under this estimator.

Table 1: Version 2 primary estimates. All eligible systems are repeatedly matched to 31608 tokens. Δmin is Voynich minus comparator; brackets give its 95% replicate interval. `OTHER` is excluded.

System	Prefix	Suffix	P/S	$\min(P,S)$	Δmin [95% interval]
Voynich	**1.333**	**1.458**	**0.918**	**1.333**	target
Arabic	1.485	0.877	1.693	0.877	0.456 [0.285, 0.644]
Latin	0.543	2.642	0.206	0.543	0.767 [0.476, 0.963]
Estonian	0.701	1.939	0.362	0.701	0.622 [0.451, 0.843]
Hebrew	0.565	1.707	0.331	0.565	0.768 [0.402, 1.000]
Finnish	0.622	1.395	0.446	0.622	0.713 [0.532, 0.891]
Hungarian	0.546	0.659	0.829	0.544	0.791 [0.660, 0.931]
Turkish	0.619	0.779	0.795	0.615	0.719 [0.548, 0.875]
Italian	0.604	0.810	0.746	0.604	0.725 [0.605, 0.871]
N. Azerbaijani	0.714	0.840	0.850	0.704	0.627 [0.416, 0.808]
Swahili	0.819	0.998	0.821	0.819	0.513 [0.399, 0.643]
Georgian	0.985	0.812	1.213	0.811	0.520 [0.377, 0.670]
Tagalog	0.467	0.615	0.759	0.467	0.864 [0.745, 0.996]
Middle English	0.338	0.400	0.845	0.336	0.994 [0.879, 1.133]
KJV English	0.259	0.094	2.755	0.094	1.234 [1.130, 1.373]

3.4 Sensitivity to `OTHER` and thresholds

Including `OTHER` in the class mean changes the Voynich medians to prefix 1.292, suffix 1.363, ratio 0.951, and minimum-side 1.291. The effect remains elevated but is smaller, which confirms that the catch-all class is a material estimator choice and should not be hidden in the headline mean.

Under the 27 declared bucket definitions, Voynich is labeled `SYMM-HIGH` in 89% to 100% of replicates, depending on thresholds. This is evidence of threshold robustness within the declared grid, not an invariant categorical identity. Continuous estimates remain primary.

3.5 Retractions and narrowed findings

The audit produced the following claim changes relevant to this paper:

- The prefix 1.52, suffix 1.54, ratio 0.99 comparison is retired. It came from a boundary-crossing, unmatched, asymmetric estimator.
- "Unique among natural languages" is retired. The defensible result is limited to 14 eligible frozen comparators and the Version 2 method.
- Canonical AIIN-family invariance is rejected. Only raw substring density has similar observed Currier A/B means, without an equivalence test.
- The claimed five-to-ninefold joint-feature compounding is rejected; a permutation estimator gives a narrower descriptive range of approximately 1.53 to 4.27.
- The claim that all five agreement cascades survive the corrected procedure is rejected; the July procedure reports four of five, and clustered inference remains open.
- A productive-morphology interpretation of edit-distance correlation remains retired because a character-trigram null reproduces it.
- Synthetic recodings are not independent non-EVA paleographic alphabets. Such validation remains unperformed.
- No single generated adversarial system has demonstrated all seven historical checklist criteria simultaneously. The checklist is not a classifier for encoded natural language.

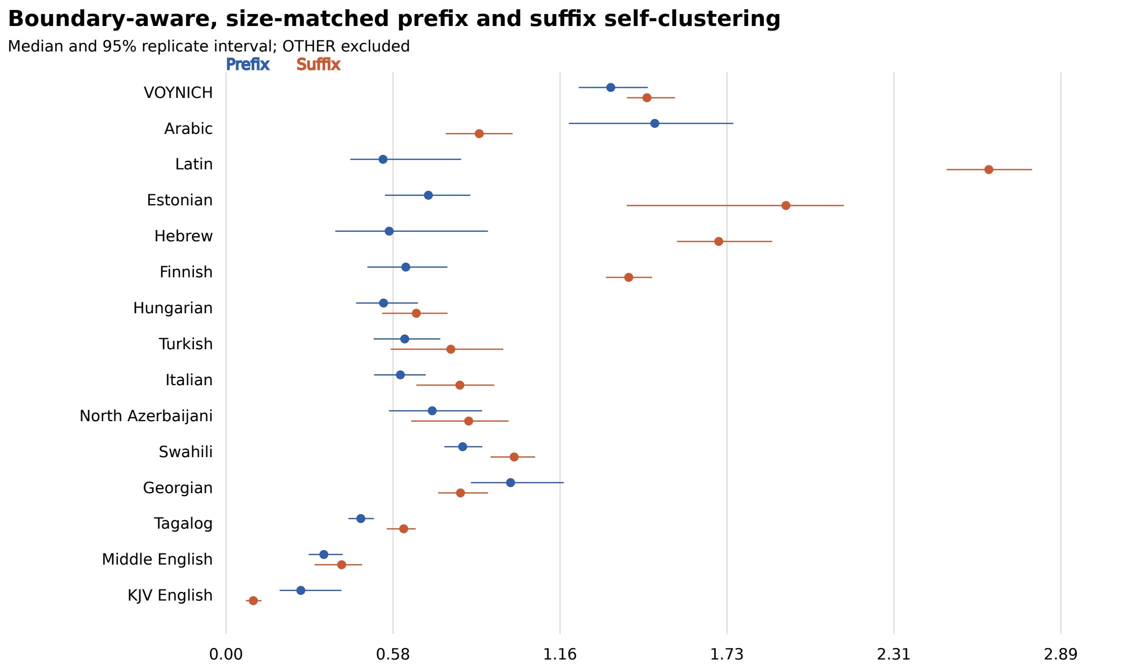

Figure 1: Programmatically generated median prefix and suffix self-clustering with 95% replicate intervals. Natural sequence boundaries are preserved and all eligible samples are matched to 31608 tokens.

These historical analyses are preserved in archival paths. They are not combined into a new omnibus conclusion.

4 Discussion

The within-line transition estimates and the matched prefix/suffix comparison show that the EVA token stream has reproducible structural regularities under declared representations. The most defensible comparative statement is that Voynich has a higher weaker-side self-clustering estimate than each of the 14 eligible comparators in the frozen set. This balances information from both token edges and avoids turning a high score on only one side into evidence of bidirectional elevation.

The result is narrower than claims in earlier versions. Comparator membership is not a random sample of the world's languages. Most comparator texts are modern Wikipedia proxies, two are historical literary works, and none is a genre-matched fifteenth-century technical manuscript. A finite-set contrast cannot justify natural-language uniqueness. Nor does balanced edge clustering identify whether the generating mechanism is linguistic, constructed, stenographic, cryptographic, hybrid, or something else.

The revised analysis also separates three kinds of uncertainty that were previously conflated. Page-block resampling describes variation associated with Voynich page composition. Repeated sentence-unit subsampling describes variation among matched portions of each frozen comparator. Threshold sensitivity describes dependence on categorical cutoffs. None of these directly measures uncertainty over transcription alphabets, unobserved languages, genre, or historical mechanisms.

5 Limitations

1. **EVA dependence.** Tokenization and family discovery operate on EVA code points. Synthetic recodings do not substitute for independent paleographic alphabets.

2. **Finite and uneven comparator set.** Fourteen comparators are eligible; most are modern Wikipedia proxies. The result cannot be generalized to all natural languages or historical text systems.

3. **Ottoman Turkish exclusion.** The preserved corpus is too small for exact matching. No conclusion about its comparative profile is reported by the Version 2 estimator.
4. **Gutenberg segmentation.** Sentence-like segmentation is deterministic and boundary-preserving but heuristic.
5. **Discovery specification.** Results depend on affix length, coverage, family count, and support rules. The threshold grid varies bucket cutoffs, not every discovery hyperparameter.
6. **Resampling interpretation.** Replicate intervals summarize the declared resampling processes, not a population of all languages or all possible transcriptions.
7. **Monte Carlo resolution.** With 200 replicates, the smallest corrected two-sided sign estimate is 0.00995.
8. **Transition dependence.** Adjacent transitions overlap and are clustered within lines and pages. Matrix-wide cluster-aware inference is still required before confirmatory claims about the whole matrix.
9. **No semantic inference.** Surface-form statistics do not assign meanings, identify a language, or demonstrate decipherment.
10. **Untested mechanisms.** Sophisticated constructed, hybrid, stenographic, cipher, and historical mechanisms have not been exhaustively generated and tested through this pipeline.

6 Conclusions

The Version 2 repository supports two bounded conclusions. First, under an explicit prefix-first EVA classifier and within-line adjacency, CHEDY→QOK is elevated and AIIN→QOK is suppressed relative to an independence baseline. Second, under identical affix discovery, preserved natural boundaries, exact size matching, and repeated block-aware sampling, Voynich has a higher minimum of prefix and suffix self-clustering than each of 14 eligible frozen comparators.

These are reproducible corpus findings, not a decipherment or language identification. Their value is to specify measurements that future transcriptions, historical comparators, and explicit generative controls can attempt to reproduce or refute.

7 Data and Code Availability

Code, frozen inputs, manifests, results, tests, and audit records are available in the public GitHub repository. The canonical command is `python run_all.py`. The locked run executed dataset validation, core within-line analysis, classifier-overlap reporting, the 200-replicate prefix/suffix estimator, pytest, and direct test execution. Both test entry points collected and passed 34 tests with zero failures, skips, or warnings. The machine-readable evidence is in `results/run_manifest.json`, `results/test_report.json`, and `results/prefix_suffix_v2.json`.

Historical papers, results, release documents, and retractions remain in clearly labeled archive directories and do not override generated canonical outputs.

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